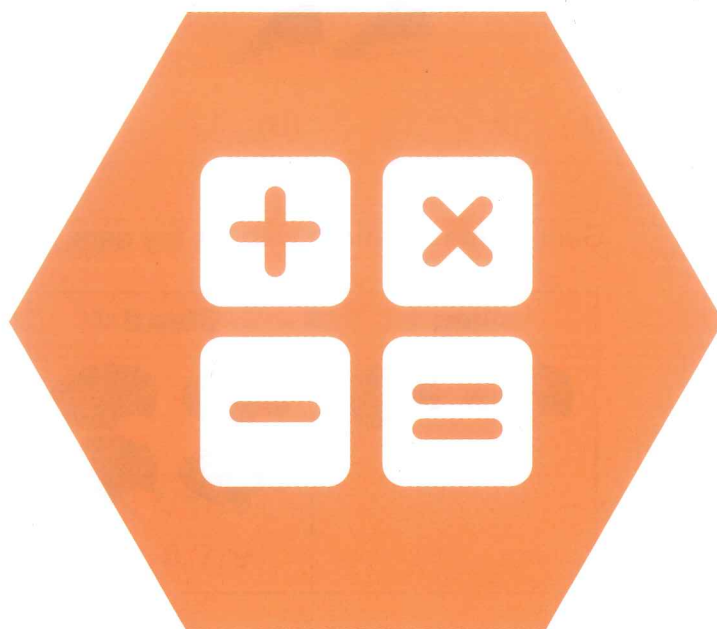




UNSW Global
AUSTRALIA

INTRODUCTORY PAPER



2015 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational Assessment Australia
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 35 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

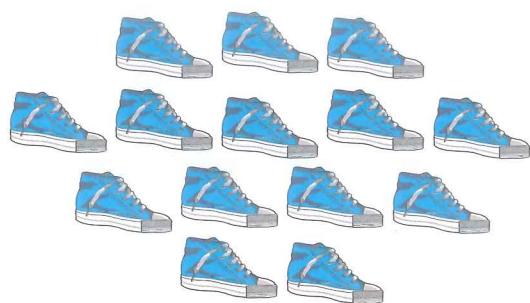
Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

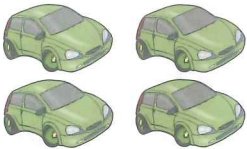
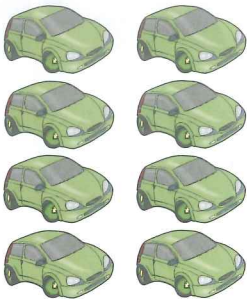
You are **NOT** allowed to use a calculator.

1. How many shoes are there in this picture?



- (A) 12 (B) 13 (C) 14 (D) 15

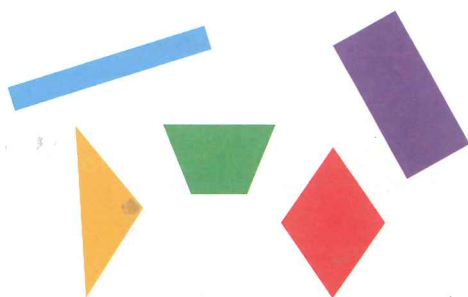
2. Sunil made a pattern with his toy cars. The table shows the steps in his pattern.

Step 1	Step 2	Step 3	Step 4
			

How many cars must there be in Step 3?

- (A) 4 (B) 5 (C) 6 (D) 8

3. How many of these shapes are rectangles?



- (A) 1 (B) 2 (C) 4 (D) 5

4. Everyone in Al's class chose one language to learn.

The table shows the languages they chose.

Language	Number of students
French	
Spanish	
Mandarin	
Japanese	

Which language was chosen by the most students?

- (A) French (B) Spanish (C) Mandarin (D) Japanese

5. Which of these has the numbers in order?

- (A) 3, 5, 7, 9, 11 (B) 3, 5, 7, 11, 9
(C) 5, 7, 9, 3, 11 (D) 11, 3, 5, 7, 9

6. Ajay wrote this number pattern.

30, 25, 20, 15, 

What number must  be?

- (A) 14 (B) 10 (C) 5 (D) 0

7. Which object is most like a cylinder?



(A)



(B)



(C)



(D)

8. Here is a picture of baby Julie.

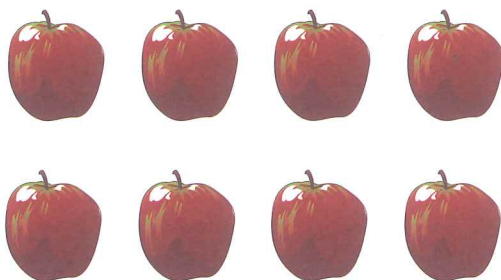


Belle is 3 times as tall as baby Julie.

How tall is Belle in centimetres (cm)?

- (A) 150 (B) 100 (C) 53 (D) 50

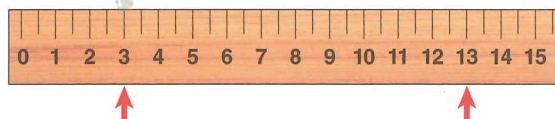
9. Ella ate one-quarter of these apples.



How many apples did she eat?

- (A) 1 (B) 2 (C) 4 (D) 6

10. What number is halfway between 3 and 13?



- (A) 7 (B) 8 (C) 9 (D) 10

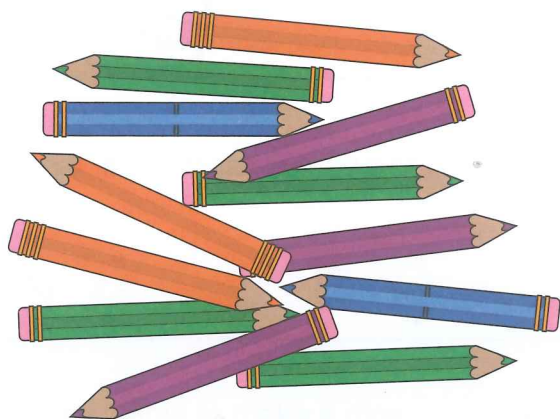
11. On Tuesday, Lisa said:



What day is her birthday?

- (A) Sunday (B) Monday (C) Tuesday (D) Wednesday

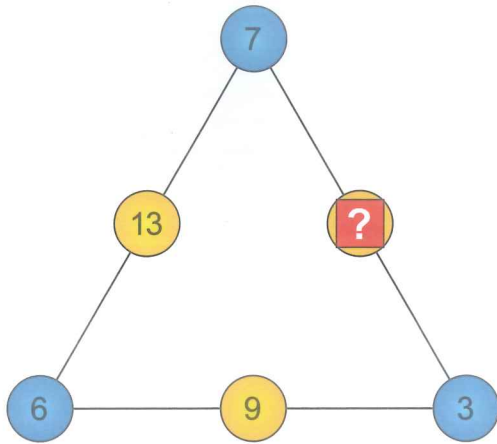
12. Chinli picked up one of these pencils without looking.



Which colour pencil was she most likely to pick up?

- (A)  (B) 
(C)  (D) 

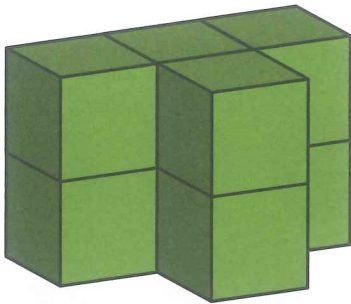
13. The numbers along each side of the triangle form a pattern.



What number must ? be?

- (A) 10 (B) 11 (C) 16 (D) 22

14. This solid is made from stacked cubes.



How many cubes make up this solid?

- (A) 14 (B) 8 (C) 7 (D) 6

15. Jane and Heidi played basketball. Each goal was worth 2 points.

Jane scored 3 goals and Heidi scored 5 goals.

How many points did they score altogether?

- (A) 8 (B) 10 (C) 15 (D) 16

16.  = 6  = 11

 +  +  +  +  = 

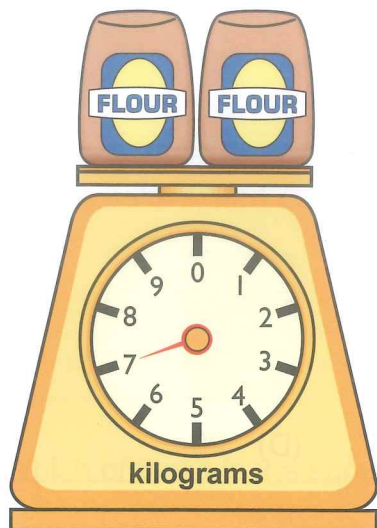
(A) 5

(B) 17

(C) 34

(D) 40

17. This picture shows the mass of 2 bags of flour.



What is the mass of 1 bag of flour, in kilograms?

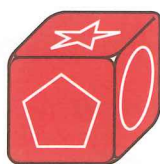
(A) $3\frac{1}{2}$

(B) $4\frac{1}{2}$

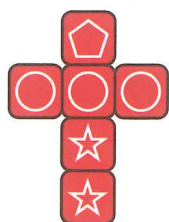
(C) 7

(D) 14

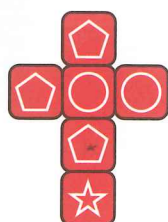
18. Jenny wants to make a dice.



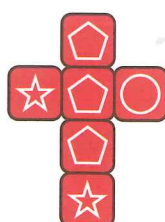
Which shape will fold to make a dice that gives Jenny the best chance of rolling a star?



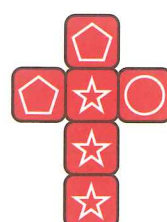
(A)



(B)



(C)



(D)

19. Joe had 13 grapes. Daniel had 8 grapes.

They ate 3 grapes each.

How many grapes did they have left in total?

- (A) 15 (B) 18 (C) 21 (D) 25

20. There were 40 students in a dance group. Each student wore only one costume. The costumes came in three colours.

Costume colour	Boys	Girls
black	4	2
gold	8	9
red	5	?

How many girls wore a red costume?

- (A) 7 (B) 11 (C) 12 (D) 28

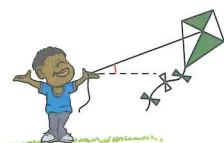
21. The diagram shows Tim flying a kite.



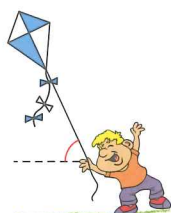
Which kite is flying at about the same angle as Tim's?



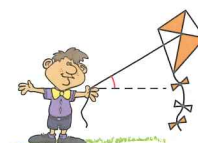
(A)



(B)



(C)

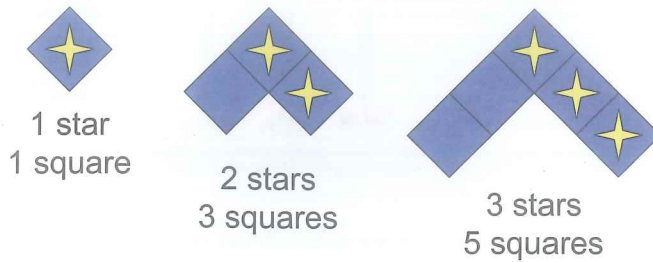


(D)

22. Which of these has the same value as $892 - 48$?

- (A) $892 - 8 + 40$
(B) $892 - 40 + 8$
(C) $892 - 50 - 2$
(D) $892 - 50 + 2$

23. Hannah drew a pattern using shapes.



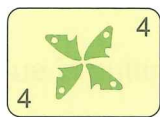
How many stars will a shape with 31 squares have?

- (A) 14 (B) 15 (C) 16 (D) 17

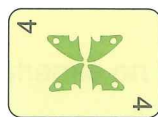
24. This is Jake's card.



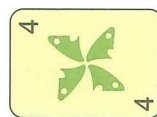
Which of these is the same card?



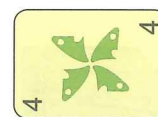
(A)



(B)



(C)

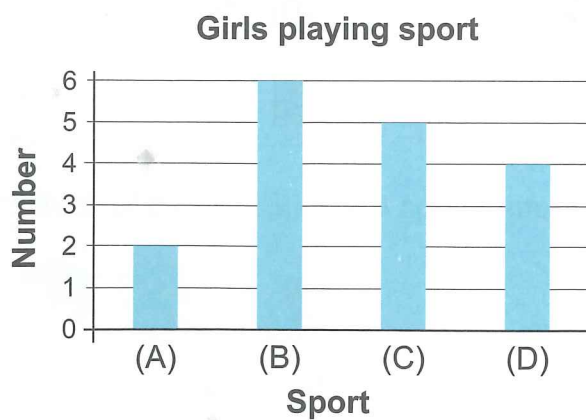


(D)

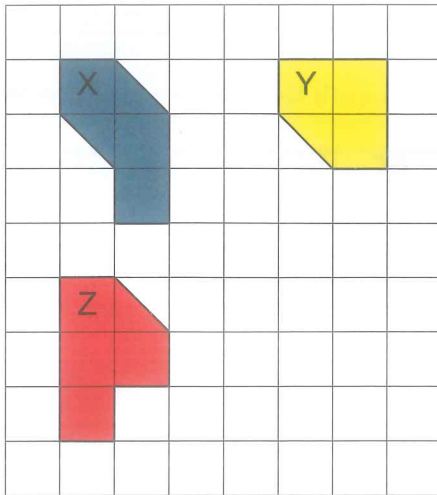
25. Each girl in Sasha's class plays only one sport.

Three more girls play tennis than play soccer.

Which column in this graph shows the girls who play tennis?



26. Hassan drew these three shapes.



Which of these lists the shapes from smallest area to largest area?

- (A) Y, X, Z (B) Y, Z, X (C) X, Z, Y (D) X, Y, Z

27. Minh stacked these boxes for a display in his shop.

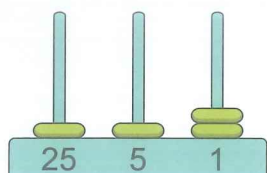
The display looks the same from the back as it does from the front.



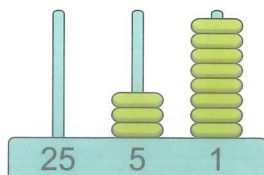
How many boxes have their faces touching only 3 other boxes?

- (A) 2 (B) 4 (C) 6 (D) 14

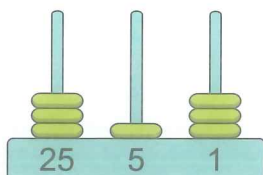
28. The beads on this abacus make up the number 32.



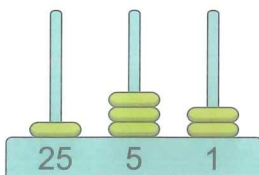
On which abacus do the beads make up the number 38?



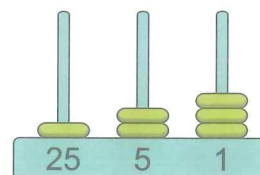
(A)



(B)



(C)



(D)

29. Brian used 30 balls of wool of equal size to make a rug. Half of the balls were green, one-tenth were white and the rest were yellow.

How many balls of yellow wool did he use?

(A) 5

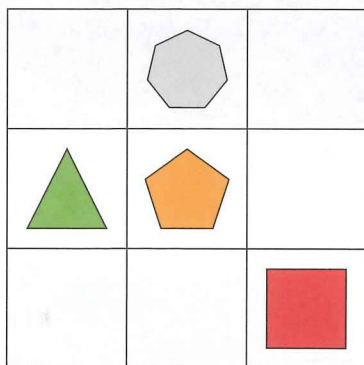
(B) 12

(C) 14

(D) 18

30. Sue is putting some shapes on a grid.

When the grid is completed, the total number of sides of all shapes in each row and in each column add up to 15.



Which of these shapes will appear most often in the completed grid?



(A)



(B)



(C)



(D)